# SEQUENCES

**Def'n** An infinite sequence of numbers is a list of numbers in a definite order

**Def'n** If $\{a_n\}$ is a sequence and $\{n_1, n_2, n_3, \ldots\}$ is a sequence of natural numbers where $n_1 < n_2 < n_3$, then $\{a_{n_1}, a_{n_2}, a_{n_3}, \ldots\} = \{a_{n_k}\}$ is a subsequence of $\{a_n\}$

**Def'n** A subsequence of $\{a_k, a_{k+1}, a_{k+2}, \ldots\}$ for some $k \in \mathbb{N}$ is a tail of $\{a_n\}$ with cutoff $k$.

**Def'n #1** $L \in \mathbb{R}$ is the limit of $\{a_n\}$ if for all $\varepsilon > 0$ there exists a $n \in \mathbb{N}$ such that if $n \geq N$, then $|a_n - L| < \varepsilon$.

If such an $L$ exists then $\{a_n\}$ <u>converges</u>. i.e.

$$\lim_{n \to \infty} a_n = L$$

If no such $L$ exists then $\{a_n\}$ <u>diverges</u>.

**Def'n #2** $\lim_{n \to \infty} a_n = L$ if for all $\varepsilon > 0$, the interval $(L - \varepsilon, L + \varepsilon)$ contains a tail of $\{a_n\}$.

**Theorem** TFAE:

1) $\lim_{n \to \infty} a_n = L$
2) For any $\varepsilon > 0$, $(L - \varepsilon, L + \varepsilon)$ contains a tail of $\{a_n\}$.
3) For any $\varepsilon > 0$, $(L - \varepsilon, L + \varepsilon)$ contains all but finitely many terms of $\{a_n\}$.
4) Every interval $(a, b)$ containing $L$ contains a tail of $\{a_n\}$.
5) Every interval $(a, b)$ containing $L$ contains all but finitely terms of $\{a_n\}$.

**Theorem** Let $\{a_n\}$ be a sequence. If it has a limit $L$ then $L$ is unique.

**Theorem** If $a_n \geq 0$ for all $n$ and $\lim_{n \to \infty} a_n = L$ then $L \geq 0$. If $\alpha \leq a_n \leq \beta$ and $\lim_{n \to \infty} a_n = L$ then $\alpha \leq L \leq \beta$.

**Def'n #1**

$\lim_{n \to \infty} a_n = \infty$ if for all $m > 0$ there exists $n \in \mathbb{N}$ such that if $n \geq N$, then $a_n > m$.

$\lim_{n \to \infty} a_n = -\infty$ if for all $m > 0$ there exists $n \in \mathbb{N}$ such that if $n \geq N$, then $a_n < m$

**Def'n #2**

Any interval $(m, \infty)$ contains a tail of $\{a_n\}$.

**Theorem**

<u>Arithmetic Rules for Limits</u>

Let $\{a_n\}$ and $\{b_n\}$ be sequences. If $\lim_{n \to \infty} a_n = L$ and $\lim_{n \to \infty} b_n = L$, then:

1) For any $C \in \mathbb{R}$, if $a_n = C$, then $C = L$
2) For any $C \in \mathbb{R}$, $\lim_{n \to \infty} C a_n = CL$
3) $\lim_{n \to \infty} (a_n + b_n) = L + M$
4) $\lim_{n \to \infty} a_n b_n = LM$
5) $\lim_{n \to \infty} \frac{a_n}{b_n} = \frac{L}{M}$, if $M \neq 0$
6) If $a_n \geq 0$ for all $n$ and $\alpha > 0$, then $\lim_{n \to \infty} a_n = L^{\alpha}$
7) For any $k \in \mathbb{N}$, $\lim_{n \to \infty} a_{n+k} = L$
8) If $\alpha > 0$ then $\lim_{n \to \infty} n^{\alpha} = \infty$
9) If $\alpha < 0$ then $\lim_{n \to \infty} n^{\alpha} = 0$

**Theorem**

If $\lim_{n \to \infty} b_n = 0$ and $\lim_{n \to \infty} \frac{a_n}{b_n} = 0$, then $\lim_{n \to \infty} a_n = 0$

**Corollary**

If $\lim_{n \to \infty} b_n = 0$ and $\lim_{n \to \infty} a_n \neq 0$, then $\lim_{n \to \infty} \frac{a_n}{b_n}$ DNE

# QUIZ #2, PRACTICE PROBLEMS #2

# PRACTICE PROBLEMS

1. (a)

Let $\varepsilon > 0$ be given.
Let $N > \frac{2}{\varepsilon} - 1$, $N \in \mathbb{N}$

Aside: $\frac{2}{N+1} < \varepsilon$
$N < \frac{2}{\varepsilon} - 1$

Then, if $n \geq N$ we get

$$\left| \frac{2n}{n+1} - 2 \right| = \left| \frac{2n - 2(n+1)}{n+1} \right| = \left| \frac{-2}{n+1} \right| = \frac{2}{n+1}$$
$$< \frac{2}{N+1} < \frac{2}{\left(\frac{2}{\varepsilon} - 1\right) + 1} = \frac{2}{\frac{2}{\varepsilon}} = \varepsilon$$

(b)

Let $\varepsilon > 0$ be given.
Let $N > \frac{1}{\sqrt{\varepsilon}} + 1$, $N \in \mathbb{N}$

Aside: $\frac{1}{(N-1)^2} < \varepsilon$
$N < \frac{1}{\sqrt{\varepsilon}} + 1$

Then, if $n \geq N$ we get

$$\left| \frac{6n - 3n^2 - 2}{(n-1)^2} + 3 \right| = \left| \frac{6n - 3n^2 - 2 + 3n^2 - 6n + 3}{(n-1)^2} \right|$$
$$= \frac{1}{(n-1)^2} < \frac{1}{(N-1)^2} = \frac{1}{\left(\left(\frac{1}{\sqrt{\varepsilon}} + 1\right) - 1\right)^2} = \frac{1}{\frac{1}{\varepsilon}} = \varepsilon$$

(c)

Let $m > 0$
Let $N > \log_2(1-m)$, $N \in \mathbb{N}$

Aside: $1 - 2^N <$
$\log_2(1-m) < N$

Then, if $n \geq N$ we get

$$1 - 2^n < 1 - 2^N < 1 - 2^{(\log_2(1-m))} = 1 - (1-m) = m$$

2. (a)

The statement is true since $a_n \to \infty$ and $b_n \to \infty$, there exists $N$ such that $a_n + b_n > m$ for all $n > N$. Then,

$a_n > m$ for all $n > N_1$
$b_n > m$ for all $n > N_2$

Choose $N = \max\{N_1, N_2\}$. Then $a_n + b_n > m$ for all $n > N$.

(b)

Let $a_n = n^2$, $b_n = n$. Then, $\lim_{n \to \infty} a_n = \lim_{n \to \infty} b_n = \infty$

but $\lim_{n \to \infty} (a_n - b_n) = n^2 - n = n(n-1) = \infty$

Hence the statement is false

(c)

Let $a_n = -1$, $b_n = 1$, $c_n = \sin(n)$. Then, $a_n \leq b_n \leq c_n$ however $\lim_{n \to \infty} c_n$ DNE.

Hence the statement is false.

3. Assume $\lim_{n \to \infty} a_n = 0$. Then for all $\varepsilon > 0$ there exists $n \geq N$ such that $|a_n| < \varepsilon$.

Choose $\varepsilon = \frac{1}{2}$. Then there exists some $N_0$ such that if $n \geq N_0$ then $|a_n| < \frac{1}{2}$.

However, for any perfect square $> 0$ $|a_n| \geq 1 > \frac{1}{2}$, a contradiction. Thus, $a_n$ does not converge to 0.

4. ?

5. (a) Since $\lim_{n \to \infty} a_n = L$ then

$$|a_n - L| < \varepsilon \Rightarrow L - \varepsilon < a_n < L + \varepsilon$$

Similarly, since $\lim_{n \to \infty} b_n = M$ then

$$|b_n - M| < \varepsilon \Rightarrow M - \varepsilon < b_n < M + \varepsilon.$$

Then, since $a_n \leq b_n$,

$$L - \varepsilon < a_n \leq b_n < M + \varepsilon \text{ as required.}$$

(b) Assume $L = m + d$. Then,

$$m + d - \varepsilon < a_n \leq b_n < m + \varepsilon$$

Choose $\varepsilon = \frac{d}{4}$. We get

$$m + \frac{3d}{4} < a_n \leq b_n < m + \frac{d}{4}$$

Since $\frac{3d}{4} \not< \frac{d}{4}$, $L \leq m$.

6. Case 1: $L = 0$

Let $\varepsilon > 0$ be given. Suppose $\lim_{n \to \infty} a_n = 0$. Then,

if $n \geq N$ there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon^2$.

$a_n < \varepsilon^2 \Rightarrow \sqrt{a_n} < \sqrt{\varepsilon^2} = \varepsilon$

Since $\sqrt{a_n} \to \sqrt{L} = 0$, the case holds.

Case 2: $L \neq 0$.

Let $\varepsilon > 0$ be given. Suppose $\lim_{n \to \infty} a_n = L$. Then,

if $n \geq N$ there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon^2$.

$$|\sqrt{a_n} - \sqrt{L}| = \frac{|a_n - L|}{\sqrt{a_n} + \sqrt{L}} < \frac{\sqrt{L}\,\varepsilon}{\sqrt{a_n} + \sqrt{L}}$$

$$< \frac{\sqrt{L}\,\varepsilon}{\sqrt{L}} = \varepsilon$$

Since $\sqrt{a_n} \to \sqrt{L}$, the case holds.

7. (a) F
(b) T
(c) T
(d) F
(e) T

8. (a) T
(b) F
(c) F
(d) T
(e) T